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A Reasoning sample of LLM

Fig. 7 shows an example of LLM reasoning process. From the figure, we can see:

1. The model's reasoning is organized as separated steps separated by "\n\n".
2. The reflection process of model, marked as red in the figure, is triggered by specific keywords like "wait" in this example.

This justifies our reflection direction extraction and intervention method.

Question:

Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder at the farmers' market daily for $2 per fresh duck egg. How much in dollars does she make every day at the farmers' market?

Thinking Process:

Okay, let me try to figure out how much Janet makes every day at the farmers' market. So, the problem says that Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder at the farmers' market daily for $2 per fresh duck egg. I need to find out how much she makes each day. Hmm, let me break this down step by step.

First, let me figure out how many eggs Janet starts with each day. The problem says she lays 16 eggs per day. So, she begins with 16 eggs. Then she eats three for breakfast every morning. So, she uses 3 eggs for breakfast. After that, she bakes muffins for her friends, using four eggs each day. So, she uses 4 eggs for baking.

Wait, so she uses 3 eggs in the morning and 4 eggs for the muffins. Let me calculate the total number of eggs she uses in a day. So, 3 plus 4 is 7 eggs. So, she uses 7 eggs each day.

She starts with 16 eggs, uses 7, so how many eggs does she have left? Let me subtract 7 from 16. 16 minus 7 is 9. So, she has 9 eggs left after using them for breakfast and baking.

Now, these remaining 9 eggs, she sells at the farmers' market. The problem says she sells each fresh duck egg for $2. So, I need to figure out how much money she makes from selling these 9 eggs.

To find the total revenue, I can multiply the number of eggs she sells by the price per egg. So, 9 eggs times $2 per egg. Let me compute that. 9 times 2 is 18. So, she makes $18 each day at the farmers' market.

Wait, let me double-check my steps to make sure I didn't make a mistake. She lays 16 eggs, uses 3 for breakfast, 4 for muffins, which totals 7 eggs used. 16 minus 7 is 9 eggs left. 9 eggs times $2 is $18. Hmm, that seems right.

I think that makes sense. So, she starts with 16, uses 7, sells 9, making her $18 each day. Yeah, that adds up.

Just to visualize it, maybe I can represent it with some equations. Let me define variables for clarity. Let E be the number of eggs laid per day, which is 16. Let B be the number of eggs she eats for breakfast, which is 3. Let M be the number of eggs she uses for muffins, which is 4. So, the total eggs used per day would be B plus M, so 3 plus 4 is 7.

Then, the eggs she has left would be E minus \(B + M\), which is 16 minus 7, which is 9. Then, the revenue R from selling these remaining eggs is 9 times the price per egg, which is 2. So, R is 9 times 2, which is 18.

Yes, that all checks out. So, I think I did it correctly. She makes $18 each day at the farmers' market.

Final Answer Janet makes \boxed{18} dollars every day at the farmers' market.

Figure 7: Sample reasoning process of DeepSeek-r1-llama-8b model on a gsm8k question.

B Reflection direction attribution to attention heads

In this section, we study the attribution of reflection direction to individual attention heads. We first take the reflection direction at self-attention layers from a DeepSeek-r1-qwen-1.5b model. Then, we compare it with the average activation of individual attention heads. We calculate the projection of average activation on the reflection direction and plot the heatmap in Fig. 8. From the figure, we can see:

1. Last layer (layer 27) has largest projection magnitude among all the layers. The reason may be this layer controls generation of keywords directly controlling reflection (e.g., “wait").
2. The heads that have positive projection on reflection direction are sparse and located mostly in deeper layers of the model. These heads may direction control model’s reflection behavior.

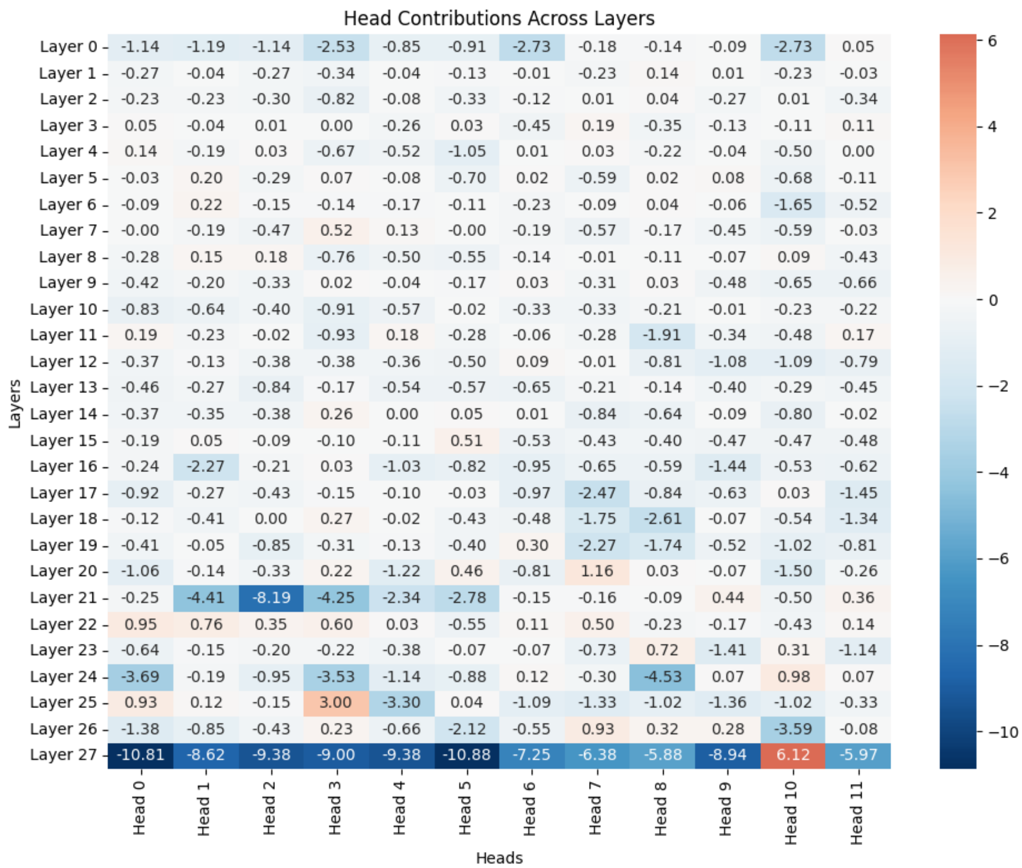

Figure 8: Projection of average activation of attention heads on the reflection direction. We use GSM8k dataset and extract results on a DeepSeek-R1 qwen 1.5b model which has 28 layers and 12 attention heads per layer.

C Study of model performance on reflection rate

In this section, we further study how the model’s performance depends on the reflection rate by categorizing the questions into three classes according to model’s accuracy on this question: Hard($acc < 50\%$), Medium ($50\% \leq acc \leq 80\%$) and Easy ($acc > 80\%$). We study the correctness vs. reflection rate for these three categories individually and show the results in Fig. 9. We show the results for base model without intervention and with intervention of -0.98. From the results, we can see: